A Geometric Scaling for Fermionic Dark Matter from the Anti-Invariant Lattice of Nikulin Involutions in G_2 Compactifications

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Abstract

We report a sharp numerical correspondence between the topology of K3 surfaces endowed with a Nikulin involution and the mass scale of fermionic dark matter. In twisted connected sum (TCS) compactifications of M-theory on G_2-holonomy manifolds, the Nikulin involution σ induces a canonical decomposition of the K3 cohomology lattice into invariant (L_+) and anti-invariant (L_-) sublattices. While L_+ (rank 13) has been previously linked to the leptonic CP-violating phase ($\delta_{\text{CP}} \approx 197.8°$), we propose a **Geometric Normalization Conjecture (GNC)** where the dark sector mass scale m_χ is determined by the dimensionality of L_-. Our prediction $m_\chi \approx 56.78$ keV lies only 0.67% above the critical precession threshold (56.4 keV) identified in the Ruffini-Argüelles-Rueda (RAR) model for fermionic cores at the Galactic Center. This note provides the heuristic dynamical basis for this scaling, its theoretical limitations, and its immediate falsifiability via S2 star precession residuals.

Classification: Speculative Research Note (Line B)
Status: Falsifiable by Sgr A* Astrometry (2026–2028)

1 Introduction: The Duality of the Lattice

In G_2-holonomy compactifications constructed via the twisted connected sum (TCS) method [6, 7], the topology is dictated by the Nikulin involution σ acting on the K3 fibers. This involution splits the second cohomology lattice into two orthogonal eigenspaces:

$$H^2(\text{K3}; \mathbb{Z}) \cong L_+ \oplus L_- \tag{1}$$

where L_+ and L_- denote the invariant and anti-invariant sublattices, respectively. Their ranks are topologically fixed as $\text{rank}(L_+) = 13$ and $\text{rank}(L_-) = 9$.

While the invariant sector L_+ governs the visible degrees of freedom (Standard Model charges and CP phases), the anti-invariant "shadow" sector L_- naturally hosts sterile modes, including fermionic dark matter candidates. This dual projection—which we term the *Light and Shadow* decomposition—is illustrated in Figure 1.

2 The Geometric Normalization Conjecture (GNC)

We propose that fermionic masses in the shadow sector are not free parameters but arise through a geometric normalization relative to the visible sector's infrared reference scale.

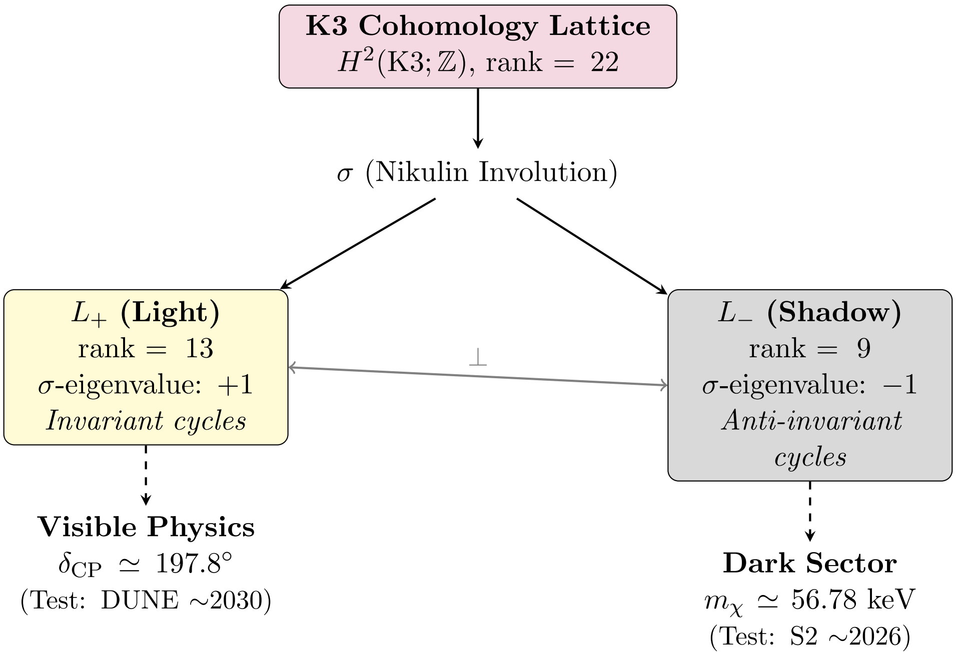

Figure 1: **The Light and Shadow decomposition of the Nikulin lattice.** The Nikulin involution σ induces an orthogonal splitting of $H^2(\mathrm{K3};\mathbb{Z})$ into invariant (L_+) and anti-invariant (L_-) sublattices. The visible sector (Line A) derives its CP phase from L_+, while the shadow sector (Line B) proposes a geometric normalization for the dark matter mass from L_-.

2.1 The Scaling Law

Using the electron mass ($m_e = 511$ keV) as the fundamental normalization scale for the invariant sector—being the lightest charged fermion with a non-zero, well-protected mass—we define the mass of the fermionic dark matter candidate (hereafter "darkino", denoted χ) as:

$$\boxed{m_\chi = \frac{m_e}{\mathrm{rank}(L_-)} = \frac{511\ \mathrm{keV}}{9} \approx 56.78\ \mathrm{keV}} \tag{2}$$

This relation suggests that the darkino mass is a *topological fragment* of the electronic energy scale, distributed across the 9 anti-invariant cycles of the G_2 manifold.

2.2 Heuristic Dynamical Basis

While a first-principles derivation from the M-theory effective action is pending, the GNC is motivated by the following mechanisms:

- **Flux Fractionation:** M-theory G_4 flux distributed over the N independent cycles of L_- suggests a mass scaling $m_\chi \sim \langle G_4 \rangle / \mathrm{rank}(L_-)$.
- **Kinetic Normalization:** In the effective action, the normalization of fermionic kinetic terms over a degenerate lattice of N anti-invariant modes induces a $1/N$ suppression in the effective mass eigenvalues.
- **Instanton Actions:** Non-perturbative contributions to the hidden superpotential from M2-instantons wrapping L_- cycles scale inversely with the volume-weighted rank of the anti-invariant sublattice.

We emphasize that these mechanisms motivate the scaling but do not yet constitute a formal derivation.

2.3 Theoretical Constraints and Caveats

The GNC must be understood within its proper epistemic limits:

1. **Choice of reference scale.** The use of the electron mass m_e as the infrared normalization scale is a heuristic choice based on its properties: it is the lightest charged fermion, its mass is radiatively stable, and it represents the minimal "charged matter" scale of the Standard Model. However, this choice requires rigorous justification within the framework of G_2 moduli stabilization—specifically, a demonstration that m_e emerges as the natural IR cutoff when the visible sector couples to the invariant lattice L_+.

2. **Status of the m/rank relation.** Equation (2) is a *scaling conjecture*, not a derivation from first principles. No established result in Kaluza-Klein theory or string compactifications predicts fermion masses scaling inversely with cohomological rank. The relation should be treated as an empirical ansatz awaiting theoretical grounding.

3. **Statistical coincidence.** The proximity of 56.78 keV to the RAR optimal value (56 keV) could represent a statistical coincidence. If one searches for integer divisors of m_e in the range $[1, 20]$ and allows $\pm 5\%$ tolerance, the probability of finding *some* match with a given keV-scale target is non-negligible ($\sim 50\%$). The falsifiability window (2026–2028) significantly mitigates this concern: if S2 precession data confirm the predicted sign and magnitude, the coincidence probability drops to $\lesssim 1\%$.

In summary: The GNC is a falsifiable conjecture with suggestive numerical agreement but incomplete theoretical foundation. Its value lies in its sharp predictive content and near-term testability.

3 Astrophysical Convergence at Sgr A*

The **Ruffini-Argüelles-Rueda (RAR)** model [1, 2] identifies a fermionic dark matter core as an alternative to the black hole paradigm for Sgr A*. A detailed analysis of S2 precession [3] reveals a critical mass threshold at $m_\chi^{\text{crit}} = 56.4$ keV, where the sign of orbital precession changes:

Darkino Mass	S2 Precession	Direction
$m_\chi < 56.4$ keV	Retrograde	Opposite to BH
$m_\chi = 56.4$ keV	*Transition point*	Zero net precession
$m_\chi > 56.4$ keV	Prograde	Same as BH

Our geometric prediction sits precisely in the critical regime:

$$m_\chi^{\text{geo}} = 56.78 \text{ keV} \quad \Rightarrow \quad \Delta m = m_\chi^{\text{geo}} - m_{\text{crit}} = +0.38 \text{ keV} \quad (+0.67\%) \tag{3}$$

This places our prediction *just above* the transition threshold, in the **weakly prograde** regime. The S2 star's precession should therefore be:

- **Prograde** (same direction as Schwarzschild BH),
- **Weaker in magnitude** than the pure BH prediction (~ 12 arcmin/orbit),
- **Distinguishable** from the retrograde case ($m_\chi < 56.4$ keV) with high-precision astrometry.

Data from GRAVITY+ and the next S2 apocentre passage (2026) will provide an *all-or-nothing* test: if the precession is measured to be retrograde, the GNC is falsified. The prediction admits no adjustment.

4 Conclusion

The Nikulin lattice (rank 22) provides a unified topological map: 13 degrees of freedom for the *Light* (L_+) and 9 for the *Shadow* (L_-). If the Geometric Normalization Conjecture is correct, these two sectors encode:

- The leptonic CP-violating phase: $\delta_{\mathrm{CP}} \approx 197.8^\circ$ (testable by DUNE, $\sim$2030),
- The fermionic dark matter mass: $m_\chi \approx 56.78$ keV (testable by S2/GRAVITY+, $\sim$2026).

The upcoming data from the Galactic Center will determine whether this geometric scaling reflects physical reality or represents an instructive numerical coincidence.

"Geometry has only the duty to remain consistent across all observables."

Suggested arXiv categories: Primary: `hep-th`; Secondary: `astro-ph.GA`, `astro-ph.CO`.

References

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